Taehun Cha

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Taehun Cha is a Ph.D. candidate in the Department of Mathematics at Korea University. His research asks how the geometry of feature learning explains why deep networks succeed on natural data — and how this understanding can guide the design of AI systems for scientific discovery.

Education

Ph.D. Candidate in Mathematics – Korea University (Advisor: Donghun Lee) Mar. 2022 — Present

- Funded by National Research Foundation (NRF) of Korea
- Dissertation: Two Faces of One Gram: Understanding Feature Learning with Virtual Covariance and Target Linearity

M.F.E. in Financial Engineering – Korea University (Advisor: Donghun Lee) Mar. 2020 — Feb. 2022

- Fully funded student
- Thesis: Understanding the Yield Curve Shift with FOMC Statements: NLP Perspective

B.A. in Sociology and Cultural Critics – Yonsei University Mar. 2012 — Aug. 2019

- Minor in Applied Statistics

Work Experience

KT Corp., Research Intern Jul. 2023 — Aug. 2023

- Focusing on hallucination problem, one of the crucial AI safety issues, built an automated pipeline to construct a hallucination dataset and a reward model to train LLM with reinforcement learning.
- Selected as an outstanding intern.

Publication

Mathematical Foundations of Artificial Intelligence

Taehun Cha, Daniel Beaglehole, Adityanarayanan Radhakrishnan, and Donghun Lee. 2026. “The Weight Gram Matrix Captures Sequential Feature Linearization in Deep Networks.” **arXiv preprint**. arXiv:2605.06258.

Taehun Cha and Donghun Lee. 2025. “ABC3: Active Bayesian Causal Inference with Cohn Criteria in Randomized Experiments.” In The 39th Annual AAAI Conference on Artificial Intelligence (**AAAI 2025, Oral**).

Taehun Cha and Donghun Lee. 2025. “Curse of Smoothness in Functional Neural Networks.” **IEEE Signal Processing Letters**. DOI: 10.1109/LSP.2025.3624683.

Taehun Cha and Donghun Lee. 2025. “Feature Learning as a Virtual Covariance Learning.” Optimization for Machine Learning Workshop at Neurips 2025 (**OPT@Neurips 2025**).

Taehun Cha and Donghun Lee. 2025. “Emergent Linear Separability of Unseen Data Points in Last-Layer Feature Space.” Workshop on High-dimensional Learning Dynamics at ICML 2025 (**HiLD@ICML 2025**).

Artificial Intelligence for Scientific Discovery

Taehun Cha and Donghun Lee. 2026. “Consequence-Guided Information Extraction for Predicting Central Bank Communication’s Effect.” **Computational Economics**. DOI: 10.1007/s10614-025-11284-6. *[Economics]*

Taehun Cha and Donghun Lee. 2024. "Evaluating Extrapolation Ability of Large Language Model in Chemical Domain." Language + Molecules Workshop at ACL 2024.**(Lang+Moles@ACL 2024)**. *[Chemistry]*

Taehun Cha*, Jaeheun Jung*, and Donghun Lee. 2022. "Noun-MWP: Math Word Problems Meet Noun Answers." In Proceedings of the 29th International Conference on Computational Linguistics **(COLING 2022)**. *[Math Education]*

Natural Language Processing & AI Safety

Chandler Smith, Marwa Abdulhai, (...), **Taehun Cha**, (...). 2025. "Evaluating Generalization Capabilities of LLM-Based Agents in Mixed-Motive Scenarios Using Concordia." The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track **(Neurips 2025)**.

Taehun Cha and Donghun Lee. 2024. "Pre-trained Language Models Return Distinguishable Probability Distributions to Unfaithfully Hallucinated Texts." Findings of the Association for Computational Linguistics: EMNLP 2024 **(EMNLP 2024 Findings)**.

Taehun Cha and Donghun Lee. 2024. "SentenceLDA: Discriminative and Robust Document Representation with Sentence Level Topic Model." In Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics **(EACL 2024, Oral)**.

Projects on Scientific Discovery

[Chemistry] **Extrapolative Chemical Property Prediction with LLM Chemical Knowledge**. 2024.

- *Scientific Question:* Can chemical knowledge of LLM help chemical property prediction of unseen compounds?
- *Answer:* LLM can predict the increase/decrease in chemical property of unseen compounds accurately using its internal knowledge.

[Seismology] **Earthquake Detecting Algorithm from Multi-Observatory Data Using Deep Learning**. 2022.

- *Scientific Question:* Can we utilize multi-observatory data for detecting and phase-picking earthquake signals?
- *Answer:* By augmenting the convolutional neural network to the state of the art transformer model, we can exploit the multi-observatory characteristic.

[Math Education] **Large Scale Deep Learning Based Algorithm for Mathematics Problem Solving**. 2021.

- *Scientific Question:* Can LLM genuinely solve the elementary school level math word problems?
- *Answer:* LLMs were especially weak at solving math problems with noun answers, so we build a dataset and a transformer based solver.

Grant

Ph.D. Fellowship. 2025. National Research Foundation of Korea (NRF). 25,000,000 KRW (~ 20,000 USD)

- Financial support for research on feature space geometry of deep neural networks.

Junior Fellow-Research Grant. 2023. Korea University. 2,000,000 KRW (~ 1,500 USD)

- Financial support for research on sentence-level topic modeling.

AI Grand Challenge. 2021. Ministry of Science and ICT. 237,500,000 KRW (~ 167,000 USD)

- Serving as a team leader in the AI Grand Challenge, secured a follow-up project to develop an AI algorithm for solving math word problems.

Awards

Concordia Contest @ Neurips 2024 - First Place out of 197 participants. 2024. Cooperative AI Foundation, Google DeepMind, MIT, UC Berkeley, and UCL.

- Designed and developed an LLM-based AI agent that balances individual reward maximization with cooperative behavior in mixed-motive scenarios (participated individually).

AI Grand Challenge - 7th Place. 2023. Ministry of Science and ICT.

- Designed and led the development of an open-domain, multi-hop, multi-modal, document-based report-generating system (team leader).

Korean AI Competition - 4th Place. 2022. Ministry of Science and ICT.

- Developed an Automatic Speech Recognition model for the Korean language and achieved 4th place out of 103 teams (team leader).

AI Grand Challenge. 2021. Ministry of Science and ICT.

- Developed an NLP model to solve elementary math word problems based on KLUE-RoBERTa, and we were selected for follow-up research (team leader).

Invited Talk and Presentations

Feature Learning as a Virtual Covariance Learning, KIAS Center for AI and Natural Sciences 2026 Winter Workshop, Korea Institute For Advanced Study, January 05, 2026. (Poster)

Understanding Feature Learning: AGOP, RFM, and Virtual Covariance Learning, Korean Mathematical Society Fall Symposium, Korean Mathematical Society, October 22, 2025.

Cooperative AI: Focusing on the Concordia Contest, Cooperative AI Talk, AI Safety Asia, Online, April 10, 2025. `https://www.youtube.com/watch?v=c4bSITHm6Ms`

Extracting Financial Knowledge from FOMC Communications with Reinforcement Learning and Natural Language Processing, The Artificial Intelligence Symposium, Natural Science Research Institute, Gangneung-Wonju National University, June 9, 2023.

Summer Research Program

Annual Summer School on Mathematical Aspects of Data Science. National University of Singapore & Institute for Mathematical Sciences. 2026. (Travel grant recipient)

KIAS Lecture Series on Computational Neuroscience and AI. Korea Institute For Advanced Study. 2025.

Machine Learning and Mathematics. Korea Institute For Advanced Study. 2025.

NIMS Summer Tutorial on Artificial Intelligence. National Institute for Mathematical Sciences. 2024.

Teaching Experience

Korea University

- Mathematics for Artificial Intelligence (2025 Fall), Teaching Fellow.
  Implemented and delivered lectures on dimension reduction and functional neural networks, integrating recent research results into coursework.

Academic Service

- Reviewer for Neurips, TMLR, ACL Rolling Review, and Neural Computing and Applications